

STA 130A: Mathematical Statistics: Brief Course

Lecture 8: Conditional expectation and Independence

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Fall 2026, UC Davis

Announcements

Homework 3: Due Thu, Oct 15, 11:59 PM PT

- Please submit on time and follow the submission instructions
- Please review the homework problems early so you have time to ask questions if needed
- Feel free to ask questions during lecture, in office hours or post them on Piazza

Midterm 1 is in class on Mon, Oct 19

- You may bring *one **hand-written** letter-sized (8.5×11 inches), double-sided sheet of paper* with formulas and brief notes
- **Calculator:** Simple (non-graphing) calculators only
- **No textbooks** or other study aids beyond the permitted note sheet and calculator
- **SDC accommodations:** Confirm scheduling with AES online

Agenda

Last time:

- Joint, marginal, and conditional PMFs
- Functions of random variables
- Linearity of expectation

Today:

- Conditional expectation
- The law of total expectation (a.k.a. total expectation theorem)
- Independence of random variables
 - Factorization of expectation
 - Variance of sums of independent random variables

Recap: Joint, marginal, and conditional PMFs

The same joint table as Lecture 7

$p_{X,Y}(x,y)$	$y = 0$	$y = 1$	$y = 2$	$p_X(x)$
$x = 0$	1/8	1/8	1/4	1/2
$x = 1$	1/4	1/8	1/8	1/2
$p_Y(y)$	3/8	1/4	3/8	1

- Inner cells: $p_{X,Y}$
- Row sums: p_X
- Column sums: p_Y

Joint vs marginal vs conditional PMFs

- **Joint:** both $X = 1$ and $Y = 2$

$$p_{X,Y}(1,2) = P(X = 1, Y = 2) = \frac{1}{8}$$

- **Marginal:** $Y = 2$, regardless of X

$$p_Y(2) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

- **Conditional:** $X = 1$, given $Y = 2$

$$p_{X|Y}(1 | 2) = \frac{p_{X,Y}(1,2)}{p_Y(2)} = \frac{1/8}{3/8} = \frac{1}{3}$$

Marginalizing: For each $X = x$, sum over (irrelevant) $Y \in \{0, 1, 2\}$

Conditioning: Keep the $Y = 2$ column (only) and divide by its total 3/8

Illustration: Building a joint PMF sequentially

Example (Short Q&A session)

Let X be the number of questions asked in a short Q&A session, and assume

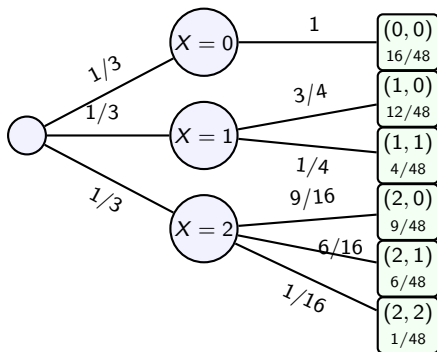
$$P(X = 0) = P(X = 1) = P(X = 2) = \frac{1}{3},$$

Suppose that given $X = x$, each question is answered incorrectly with probability $1/4$, independently of the others. Let Y be the number answered incorrectly. Thus

$$Y \mid X = x \sim \text{Binomial}(x, 1/4),$$

and therefore,

$$p_{X,Y}(x, y) = p_X(x) p_{Y|X}(y \mid x).$$



$p_{X,Y}(x, y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	$16/48$	0	0
$x = 1$	$12/48$	$4/48$	0
$x = 2$	$9/48$	$6/48$	$1/48$

Conditional expectation

Definition

The **conditional expectation** of X given an event A with $P(A) > 0$ is

$$\mathbb{E}[X | A] = \sum_x x p_{X|A}(x), \quad p_{X|A}(x) = P(X = x | A).$$

The conditional expectation of X given a value y of Y , with $p_Y(y) > 0$, is

$$\mathbb{E}[X | Y = y] = \sum_x x p_{X|Y}(x | y).$$

Example

$p_{X,Y}(x, y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	1/8	1/8	1/4
$x = 1$	1/4	1/8	1/8

From the joint PMF table

$$\mathbb{E}[X | Y = 2] = 0 \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} = \frac{1}{3}.$$

Law of total expectation

Theorem

If $A_1, \dots, A_n$ form a partition of the sample space and $P(A_i) > 0$ for all i , then

$$\mathbb{E}[X] = \sum_{i=1}^n P(A_i) \mathbb{E}[X | A_i]$$

Proof. For each x ,

$$p_X(x) = \sum_{i=1}^n P(A_i) p_{X|A_i}(x),$$

by the PMF version of the law of total probability. Therefore,

$$\begin{aligned} \mathbb{E}[X] &= \sum_x x p_X(x) = \sum_x x \sum_{i=1}^n P(A_i) p_{X|A_i}(x) = \sum_{i=1}^n P(A_i) \sum_x x p_{X|A_i}(x) \\ &= \sum_{i=1}^n P(A_i) \mathbb{E}[X | A_i]. \end{aligned}$$

Law of total expectation (cont'd)

Theorem

For a partition $A_1, \dots, A_n$ with $P(A_i) > 0$ for all i ,

$$\mathbb{E}[X] = \sum_{i=1}^n P(A_i) \mathbb{E}[X \mid A_i].$$

Two useful forms of the same rule:

- For discrete random variables X and Y ,

$$\mathbb{E}[X] = \sum_y p_Y(y) \mathbb{E}[X \mid Y = y]$$

- For any event B with $P(A_i \cap B) > 0$ for all i ,

$$\mathbb{E}[X \mid B] = \sum_{i=1}^n P(A_i \mid B) \mathbb{E}[X \mid A_i \cap B]$$

Law of total expectation: Toy example

Example

Let X be uniform on $\{0, 1, 2\}$. Given $X = x$, a student guesses independently and uniformly on x questions, each with four choices and exactly one correct answer. Let Y count correct answers. Then

$$Y \mid X = x \sim \text{Binomial}(x, 1/4),$$

so

$$\mathbb{E}[Y \mid X = x] = \frac{x}{4}.$$

Therefore,

$$\begin{aligned}\mathbb{E}[Y] &= \sum_x \mathbb{E}[Y \mid X = x] p_X(x) \\ &= \sum_x \frac{x}{4} p_X(x) = \frac{1}{4} \mathbb{E}[X] \\ &= \frac{1}{4} \cdot \frac{0 + 1 + 2}{3} = \frac{1}{4}.\end{aligned}$$

Exercise for self-study: Portfolio example (1/4)

Example (State-dependent investment returns)

Suppose the market state after one year is $M \in \{\text{boom, normal, crash}\}$, with probabilities

$$P(M = \text{boom}) = 0.20, \quad P(M = \text{normal}) = 0.75, \quad P(M = \text{crash}) = 0.05.$$

Consider two one-year investment options with state-dependent returns:

Market state	boom	normal	crash
Stock fund (R_S)	30%	10%	-40%
Hedge option (R_H)	-10%	-5%	100%

Assume returns are *fixed within each market state*. Thus the table gives the conditional means:

$$R_S(m) = \mathbb{E}[R_S \mid M = m], \quad R_H(m) = \mathbb{E}[R_H \mid M = m].$$

By the total expectation theorem,

$$\mathbb{E}[R_S] = 0.20 \times 30\% + 0.75 \times 10\% + 0.05 \times (-40\%) = 11.5\%,$$

$$\mathbb{E}[R_H] = 0.20 \times (-10\%) + 0.75 \times (-5\%) + 0.05 \times (100\%) = -0.75\%.$$

Exercise for self-study: Portfolio example (2/4)

Example (Maximizing expected return)

Suppose you invest a fraction $w \in [0, 1]$ in the stock fund and the remaining $1 - w$ in the hedge.

The portfolio return is

$$R_w = wR_S + (1 - w)R_H.$$

By linearity of expectation,

$$\mathbb{E}[R_w] = w \mathbb{E}[R_S] + (1 - w) \mathbb{E}[R_H] = w \times 11.5\% + (1 - w) \times (-0.75\%).$$

Equivalently,

$$\mathbb{E}[R_w] = 12.25\% w - 0.75\%.$$

Since this is increasing in w , the expected return is maximized at

$$w = 1.$$

Thus, if the goal is to maximize expected return *alone*, the best choice is to invest everything in the stock:

$$\max_{0 \leq w \leq 1} \mathbb{E}[R_w] = 11.5\%.$$

Exercise for self-study: Portfolio example (3/4)

Example (A simple hedging strategy)

Now suppose instead that you want the portfolio return to be *nonnegative in every market state*. Observe

$$R_w = wR_S + (1 - w)R_H = \begin{cases} 40\% w - 10\%, & \text{boom,} \\ 15\% w - 5\%, & \text{normal,} \\ 100\% - 140\% w, & \text{crash.} \end{cases}$$

To keep the return nonnegative in every state, we need

$$40\% w - 10\% \geq 0, \quad 15\% w - 5\% \geq 0, \quad 100\% - 140\% w \geq 0.$$

These inequalities imply

$$w \geq \frac{1}{4}, \quad w \geq \frac{1}{3}, \quad w \leq \frac{5}{7}, \quad \text{and thus,} \quad \frac{1}{3} \leq w \leq \frac{5}{7}$$

Since $\mathbb{E}[R_w] = 12.25\% w - 0.75\%$ is increasing in w , the best hedged portfolio is at $w = \frac{5}{7}$.

Exercise for self-study: Portfolio example (4/4)

Example (A simple hedging strategy)

The portfolio with $w = \frac{5}{7}$ gives state-by-state returns

$$R_{5/7} = \begin{cases} \frac{13}{70} \approx 18.57\%, & \text{boom,} \\ \frac{2}{35} \approx 5.71\%, & \text{normal,} \\ 0, & \text{crash,} \end{cases}$$

and expected return

$$\mathbb{E}[R_{5/7}] = 12.25\% \times \frac{5}{7} - 0.75\% = 8\%.$$

Takeaway: The law of total expectation computes expected returns from market states. A nontrivial portfolio-allocation problem appears once we add a hedging/risk constraint.

Pop-up quiz

Which identity correctly expresses the law of total expectation for discrete random variables X and Y ?

- A) $\mathbb{E}[X \mid Y = y] = \mathbb{E}[X] p_Y(y)$
- B) $\mathbb{E}[X] = \mathbb{E}[Y \mid X = x]$
- C) $\mathbb{E}[X] = \sum_y \mathbb{E}[X \mid Y = y]$
- D) $\mathbb{E}[X] = \sum_y \mathbb{E}[X \mid Y = y] p_Y(y)$

Answer: D.

The unconditional mean is the weighted average of the conditional means, with weights $p_Y(y)$.

Follow-up: Why do we need the weights $p_Y(y)$?

Independence of a random variable from an event

Definition

Let A be an event with $P(A) > 0$. We say that X is **independent of** A if

$$p_{X|A}(x) = p_X(x), \quad \forall x.$$

Equivalently, for every x , the event $\{X = x\}$ is independent of A .

Example (Two fair die rolls)

Let X be the first roll of two independent fair dice, and let

$$A = \{\text{the sum of the two rolls is even}\}.$$

For each $x \in \{1, \dots, 6\}$, exactly three values of the second roll make the sum even, so

$$P(X = x | A) = \frac{3/36}{18/36} = \frac{1}{6} = p_X(x).$$

Independence of random variables

Definition

Discrete random variables X and Y are **independent** if

$$p_{X,Y}(x,y) = p_X(x)p_Y(y), \quad \forall (x,y).$$

Equivalently, for every y with $p_Y(y) > 0$,

$$p_{X|Y}(x | y) = p_X(x), \quad \forall x.$$

- Knowing the value of Y does not change the distribution of X .
- The condition must hold for every pair (x,y) , not just some of them.
- If X and Y are independent, then $g(X)$ and $h(Y)$ are independent for any functions g and h .

Example (A non-example)

For two independent fair die rolls, let X be the first result and Y their sum. Then $P(X = 6, Y = 12) = P(\{(6,6)\}) = \frac{1}{36}$, whereas $P(X = 6)P(Y = 12) = \frac{1}{6} \cdot \frac{1}{36} = \frac{1}{216}$. Since these are not equal, X and Y are not independent.

Example: Same experiment, yet independent

Example (Two independent fair die rolls)

Let

$$X = \mathbf{1}_{\{\text{first roll is even}\}}, \quad Y = \mathbf{1}_{\{\text{sum of the two rolls is even}\}}.$$

Then $X, Y \in \{0, 1\}$, and the joint PMF^a is

$p_{X,Y}(x,y)$	$y = 0$	$y = 1$
$x = 0$	1/4	1/4
$x = 1$	1/4	1/4

Since $p_X(0) = p_X(1) = 1/2$ and $p_Y(0) = p_Y(1) = 1/2$,

$$p_{X,Y}(x,y) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = p_X(x)p_Y(y) \quad \text{for all } x, y \in \{0, 1\}.$$

So X and Y are independent.

^aEven though Y depends on both rolls, the parity of the sum is still equally likely to be even or odd once the parity of the first roll is fixed.

Pop-up quiz

Two independent fair die rolls are performed. Define

$$X = \text{first roll}, \quad A = \{\text{the second roll is even}\}.$$

Question: Which statement is correct?

- A) X is not independent of A , because they come from the same experiment.
- B) X is independent of A , and $p_{X|A}(x) = 1/6$ for all $x \in \{1, \dots, 6\}$.
- C) X is independent of A , but only for $x \in \{1, \dots, 5\}$.
- D) X is independent of A only because the dice are fair.

Answer: B.

The rolls are independent, so conditioning on the second roll being even does not change the first-roll distribution. It remains uniform on $\{1, \dots, 6\}$.

Follow-up: Would the same conclusion hold if $A = \{\text{sum is } 8\}$?

Independence implies expectation factorization

Theorem

If X and Y are independent, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

More generally, for any functions g and h ,

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)] \mathbb{E}[h(Y)].$$

Proof:

$$\begin{aligned}\mathbb{E}[g(X)h(Y)] &= \sum_{x,y} g(x)h(y) p_{X,Y}(x,y) = \sum_{x,y} g(x)h(y) p_X(x)p_Y(y) \\ &= \left(\sum_x g(x)p_X(x) \right) \left(\sum_y h(y)p_Y(y) \right) \\ &= \mathbb{E}[g(X)] \mathbb{E}[h(Y)].\end{aligned}$$

Example: Expectation factorizes with independence

Example (A binomial second moment)

Let $X_1, \dots, X_n$ be independent Bernoulli(p) random variables, and let

$$S = X_1 + \dots + X_n.$$

Then

$$S^2 = \sum_{i=1}^n X_i^2 + 2 \sum_{1 \leq i < j \leq n} X_i X_j.$$

Taking expectations gives

$$\mathbb{E}[S^2] = \sum_{i=1}^n \mathbb{E}[X_i^2] + 2 \sum_{1 \leq i < j \leq n} \mathbb{E}[X_i X_j].$$

Since $X_i^2 = X_i$ and the X_i are independent,

$$\mathbb{E}[X_i^2] = p, \quad \mathbb{E}[X_i X_j] = \mathbb{E}[X_i] \mathbb{E}[X_j] = p^2 \quad (i \neq j).$$

Therefore,

$$\mathbb{E}[S^2] = np + n(n-1)p^2.$$

Caution: Factorization alone does not imply independence

If X, Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$

However, the identity $\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$ does *not* by itself imply independence

Example

Let X be uniform on $\{-1, 0, 1\}$, and let $Y = X^2$. Then

$$\mathbb{E}[X] = 0, \quad \mathbb{E}[Y] = \frac{2}{3}, \quad \mathbb{E}[XY] = \mathbb{E}[X^3] = 0.$$

So

$$\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y].$$

But $P(Y = 0 \mid X = 0) = 1 \neq 1/3 = P(Y = 0)$, so X and Y are not independent

Factorization of $\mathbb{E}[XY]$ is a consequence, but not a characterization of independence

Independence implies variance additivity

Theorem

If X and Y are independent, then

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Proof:

$$\begin{aligned}\text{Var}(X + Y) &= \mathbb{E}[(X + Y)^2] - (\mathbb{E}[X] + \mathbb{E}[Y])^2 \\ &= \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] - \mathbb{E}[X]^2 - 2\mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y]^2 \\ &= \text{Var}(X) + \text{Var}(Y) + 2(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]).\end{aligned}$$

Noting that if X and Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ completes the proof.

More generally, for mutually independent $X_1, \dots, X_n$,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

Example: Variance of a binomial random variable

Example (Expectation and variance of a binomial random variable)

Let $X_1, \dots, X_n$ be independent Bernoulli(p) random variables, and let

$$X = X_1 + \dots + X_n.$$

Since

$$\text{Var}(X_i) = p(1 - p),$$

the variance-addition rule gives

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) = np(1 - p).$$

So for a binomial random variable $X \sim \text{Binomial}(n, p)$,

$$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1 - p).$$

Example: Mean and variance of the sample mean

Example (Sample mean of i.i.d. observations)

Let $X_1, \dots, X_n$ be independent and identically distributed, with

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2.$$

Define the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then

$$\mathbb{E}[\bar{X}_n] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mu, \quad \text{and} \quad \text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{\sigma^2}{n}.$$

- Averaging preserves the mean.
- Averaging reduces the variance by a factor of n .
- This is one mathematical reason why averaging stabilizes noisy observations.

Reminder: Same marginals, different dependence

Example (Same marginals, different dependence)

Let $X, Y \in \{0, 1\}$, and suppose both are marginally Bernoulli(1/2).

Model 1: independent

$p_{X,Y}(x, y)$	$y = 0$	$y = 1$
$x = 0$	1/4	1/4
$x = 1$	1/4	1/4

Model 2: perfectly dependent

$p_{X,Y}(x, y)$	$y = 0$	$y = 1$
$x = 0$	1/2	0
$x = 1$	0	1/2

In both models,

$$\mathbb{E}[X] = \mathbb{E}[Y] = \frac{1}{2}, \quad \mathbb{E}[X + Y] = 1.$$

But

$$\text{Var}(X + Y) = \frac{1}{2} = \text{Var}(X) + \text{Var}(Y) \quad \text{in Model 1}, \quad \text{Var}(X + Y) = 1 \quad \text{in Model 2}.$$

- Marginal PMFs alone do not determine the dependence structure.
- Dependence can significantly change quantities such as $P(X = Y)$, $\mathbb{E}[XY]$, and $\text{Var}(X + Y)$.

Pop-up quiz

Suppose X and Y are both marginally Bernoulli($1/2$).

Model 1: X and Y are independent.

Model 2: $Y = X$.

Question: Which quantity is the same in both models?

- A) $\mathbb{E}[XY]$
- B) $\text{Var}(X + Y)$
- C) $P(X = Y)$
- D) $\mathbb{E}[X + Y]$

Answer: D.

In both models, $\mathbb{E}[X] = \mathbb{E}[Y] = 1/2$, so $\mathbb{E}[X + Y] = 1$; the other quantities depend on the dependence structure.

Follow-up: What extra information beyond the marginal PMFs is needed to determine $\text{Var}(X + Y)$?

Wrap-up

Conditional expectation

- $\mathbb{E}[X \mid Y = y]$ is the average of X under the conditional PMF
- The total expectation theorem states

$$\mathbb{E}[X] = \sum_{y: p_Y(y) > 0} \mathbb{E}[X \mid Y = y] p_Y(y).$$

Independence and expectations

- Independence means $p_{X,Y}(x, y) = p_X(x)p_Y(y)$ for every (x, y)
- Under independence, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$

Sums of independent random variables

- Means add without independence; variances add for independent variables
- For $S \sim \text{Binomial}(n, p)$: $\mathbb{E}[S] = np$, $\text{Var}(S) = np(1 - p)$

Suggested reading: [BT08, Ch. 2.6–2.7]

Next lecture: Continuous distributions: PDFs, CDFs, uniform and exponential models

References



Dimitri Bertsekas and John N Tsitsiklis.

Introduction to probability, volume 1.

Athena Scientific, 2nd edition, 2008.